Laxmi Charitable Trust's

Sheth L.U.J. College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc. IT Semester V)

INTERNET OF THINGS (IoT)

IOT PRACTICAL / PROJECT

Display Message on 16×2 LCD using Raspberry Pi

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1. Project Title

Display Message on a 16×2 LCD using Raspberry Pi and the RPLCD Python library.

2. Aim

To connect a 16×2 I2C LCD module to a Raspberry Pi and display messages using a Python program.

3. Hardware Requirements

- Raspberry Pi with Raspberry Pi OS
- 16×2 LCD display with I2C backpack (PCF8574)
- Jumper wires
- Raspberry Pi power supply
- Monitor/keyboard or remote terminal (as available)

4. Software Requirements

- Raspberry Pi OS
- Python 3
- RPLCD Python library
- I2C tools and SMBus support
- Nano text editor or another Python editor

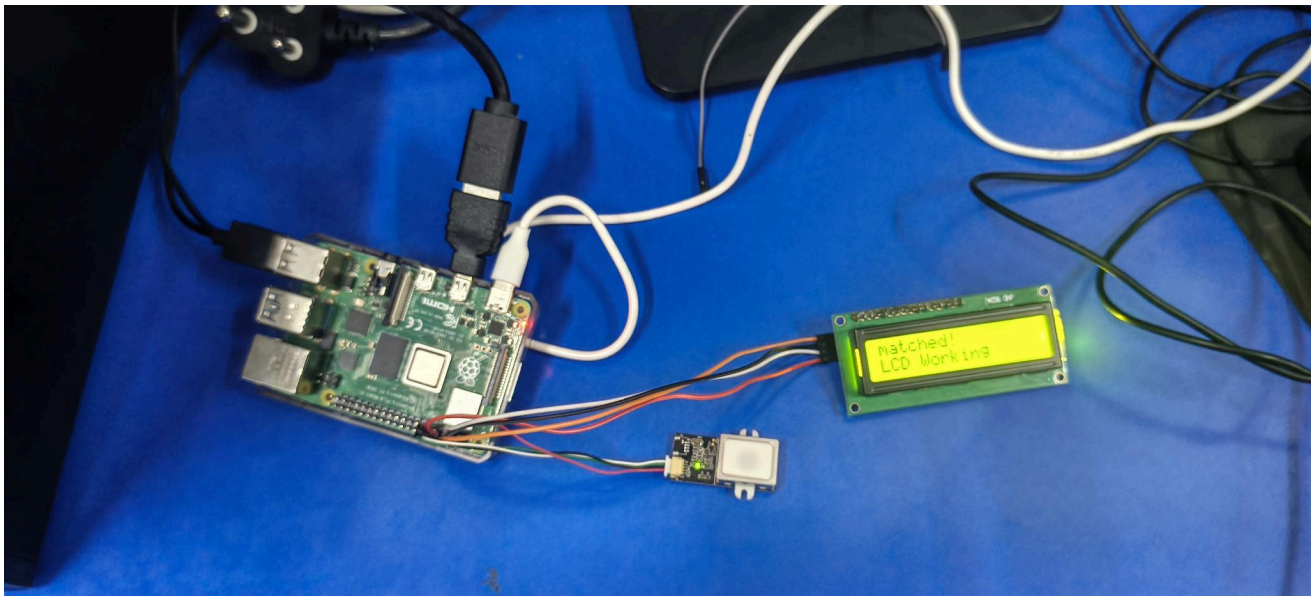
5. Procedure

Step 1: Connect the LCD to the Raspberry Pi

Connect the LCD's I2C backpack to the Raspberry Pi. The usual I2C connections are:

- VCC → 5V (physical pin 2 or 4, if supported by your module)
- GND → GND (for example, physical pin 6)
- SDA → GPIO 2 / SDA1 (physical pin 3)
- SCL → GPIO 3 / SCL1 (physical pin 5)

Check the pin labels and your LCD module's voltage requirements before powering the circuit. The hardware photo in the original PDF shows the LCD connected to the Raspberry Pi.



Step 2: Enable I2C

Open the Raspberry Pi configuration tool:

sudo raspi-config

Choose Interface Options → I2C → Enable. Reboot if the system asks you to.

sudo reboot

Step 3: Install the required packages

Run the following commands in the Terminal. The first command updates the package list; the second installs I2C tools and Python support.

```
sudo apt update
```

```
sudo apt install -y i2c-tools python3-smbus python3-venv
```

Check whether the LCD's I2C address is detected:

```
i2cdetect -y 1
```

The screenshot code uses address 0x3F. Use the address shown by your own scan (commonly 0x27 or 0x3F).

Step 4: Create a virtual environment and install RPLCD

Recent Raspberry Pi OS versions may block system-wide pip installation. Use a virtual environment instead:

```
python3 -m venv --system-site-packages ~/lcd-env
```

```
source ~/lcd-env/bin/activate
```

```
python -m pip install RPLCD
```

If the virtual-environment command fails because venv is missing, install the package with the following command, then repeat the commands above:

```
sudo apt install -y python3-full python3-venv
```

Step 5: Create the Python file

Open the Nano editor and create the file:

```
nano lcd_hello.py
```

Type or paste the following program. This version displays the same two lines as the output photo:

```
from RPLCD.i2c import CharLCD
import time

lcd = CharLCD(
    i2c_expander='PCF8574',
    address=0x3F,
    port=1,
    cols=16,
    rows=2,
    auto_linebreaks=False
)

try:
    lcd.clear()
    lcd.write_string("Hello World!")
    lcd.cursor_pos = (1, 0)
    lcd.write_string("LCD Working")
    time.sleep(30)
finally:
    lcd.clear()
    lcd.close()
```

Save in Nano: press Ctrl + O, press Enter, then press Ctrl + X to exit.

Step 6: Run the program

If the virtual environment is not active, activate it first. Then run:

```
source ~/lcd-env/bin/activate
```

```
python lcd_hello.py
```

The LCD should display “Hello World!” on the first row and “LCD Working” on the second row for about 30 seconds, then clear the display.

6. Screenshots / Observations

Figure 2 shows the Python code entered in the Nano editor.

Figure 2: Python code in Nano editor

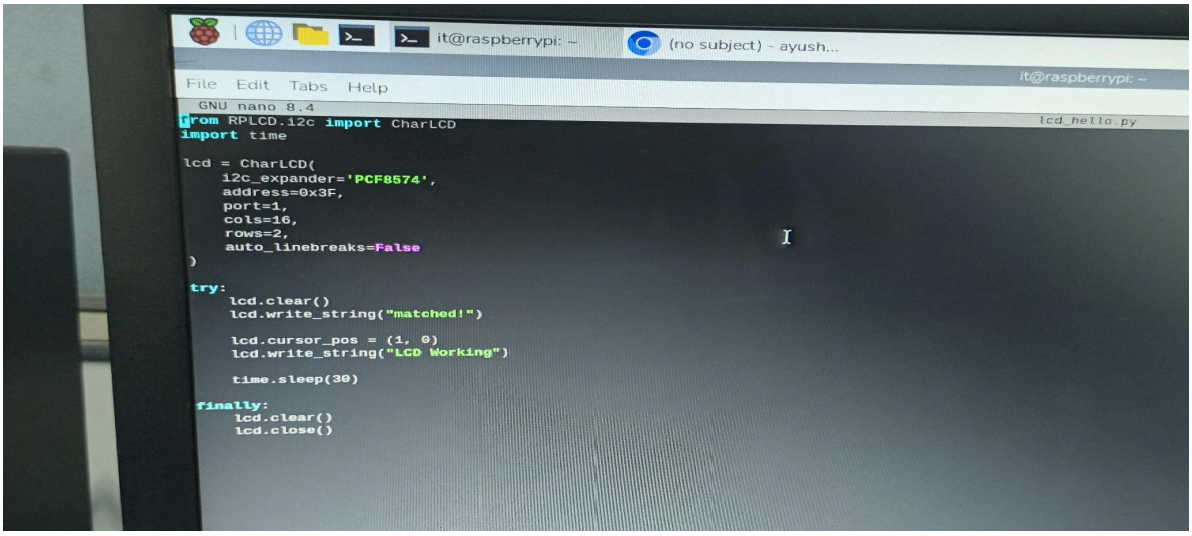
Figure 3 shows the LCD output after running the program.

Figure 3: LCD displays Hello World! and LCD Working

Figure 4 shows the Terminal and the troubleshooting attempts made during setup.

Figure 4: Terminal setup and error messages

Note: The original PDF contains the photographs/screenshots. This Word version preserves their captions and the written content.



Python code



LCD displays Hello World! and LCD Working

7. Troubleshooting Notes

- "externally-managed-environment": do not force-install packages into the system Python. Create and activate the virtual environment shown above.
- "No such file or directory" for activate: check that the virtual environment was created successfully and use the exact path ~/lcd-env/bin/activate.
- "Failed to read data from sensor": verify SDA/SCL/VCC/GND wiring, enable I2C, check the detected address with i2cdetect -y 1, and make sure the correct address is used in the code.
- If the display shows no text, adjust the small contrast potentiometer on the I2C backpack and re-check the wiring.

8. Result

The Raspberry Pi was connected to a 16×2 I2C LCD, and a Python program using the RPLCD library was prepared to display “Hello World!” and “LCD Working” on the screen.

9. Precautions

1. Switch off the Raspberry Pi before changing the wiring.
2. Verify VCC, GND, SDA and SCL connections before powering on.
3. Use the correct I2C address detected by i2cdetect.
4. Do not force-install Python packages into the system-managed environment.
5. Ensure the LCD module and wiring are compatible with the Raspberry Pi.û